

CertiFlow AutoFill Platform

Master Project Report, Developer Specifications Audit & Operational User Manual

LIVE PRODUCTION WEB APPLICATION

<https://certiflow-ten.vercel.app/>

VERSION & RELEASE

V1.0.0 Production Release

TARGET ORGANIZATIONS

Proforma & Proskills Institut

COMPLIANCE STANDARDS

RNCP / RS & Qualiopi Certification

CORE AI ENGINE

Anthropic Claude 3.5 Sonnet

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1. Executive Summary & Project Overview

1.1 Executive Summary

CertiFlow AutoFill is an enterprise-grade document automation, candidate evaluation, and certificate generation platform built specifically for French professional training centers and certification authorities (specifically **Proforma Institut** and **Proskills Institut**). Operating on live data ingested exclusively from the **AUTOMATISATION** tab of official EDOF spreadsheet registers (EDOF.xlsx / Developer of EDOF_restructure_v9.xlsx), CertiFlow automates candidate completeness validation, seed-reproducible pedagogical evaluation, mark threshold verification, multi-template Word (.docx) content-control assembly, high-fidelity PDF rendering, and automated cloud backup.

By integrating Artificial Intelligence (Anthropic Claude 3.5 Sonnet / Claude AI Engine), CertiFlow eliminates manual data copy-paste errors, ensuring that candidate portfolios, evaluation sheets (*Grille d'Évaluation*), jury transcripts (*Procès-Verbal de Jury*), and completion certificates (*Attestations de Formation*) are generated with 100% accuracy and full compliance with French RNCP (Répertoire National des Certifications Professionnelles) and RS (Répertoire Spécifique) regulations.

1.2 Key Objectives & Value Proposition

- **Single Source Data Ingestion:** Ingests candidate data exclusively from the **AUTOMATISATION** tab of EDOF.xlsx, ignoring legacy direct tabs.
- **Completeness Detection:** Pure TypeScript engine checking candidate identity, session dates, contact info, and experience background (PRET_POUR_GENERATION).
- **Dual Pipeline Support:** Automatically classifies candidates into commercial track (PRET_GENERATION_CLASSIQUE) or government track (PRET_GENERATION_WEDOF).
- **Correlated Pedagogical Chain:** Seed-reproducible scoring linking Needs Assessment (*Recueil*) → Placement Test → Eligibility → Presentation File → Evaluation Grid → PV de Jury.
- **Guaranteed Pass Floor:** Enforces minimum pass floor (≥ 10/20 or ≥ 30/60) with decision **"ADMIS"** (never "Ajourné").
- **Hardcoded Regulatory Jury Rules:** Auto-enforces regulatory jury composition for Proforma Institut and Proskills Institut.
- **Hybrid PDF Engine:** Local libreoffice-convert generation in development with automatic fallback to CloudConvert REST API on Vercel.

1.3 System Features & Capabilities Matrix

FEATURE AREA	CAPABILITIES & TECHNICAL DESCRIPTION	BUSINESS IMPACT
EDOF Data Parsing	Auto-detects and extracts candidate rows directly from the AUTOMATISATION sheet of official EDOF Excel spreadsheets.	Zero manual data entry; instant extraction.
AI Quality Audit	Invokes Anthropic Claude AI to evaluate grade distributions, attendance ratios, and compliance flags across candidate records.	Proactive error detection before document printing.
Dual Track Pipeline	Classifies candidates automatically into PRET_GENERATION_CLASSIQUE or PRET_GENERATION_WEDOF tracks.	Seamless management of commercial & government tracks.
DocxTemplater Engine	Populates up to 80 verified Word templates with dynamic XML placeholders, jury tables, and conditional text.	High-speed, standardized document assembly.

FEATURE AREA	CAPABILITIES & TECHNICAL DESCRIPTION	BUSINESS IMPACT
Hybrid PDF Engine	Uses fast local libreoffice-convert in dev/local environments with auto-fallback to CloudConvert REST API on Vercel.	Flexible, cost-effective PDF output everywhere.
Google Drive Sync	Direct OAuth service account integration uploading output packages directly into shared Google Drive folders.	Instant cloud archive & team collaboration.

2. Requirement-by-Requirement Specifications Compliance Audit

2.1 Developer Specifications Compliance Audit Matrix

SPEC §	REQUIREMENT DESCRIPTION	STATUS	IMPLEMENTATION DETAIL
§ 2 & 3	Single Source Data Ingestion (AUTOMATISATION tab in ED0F.xlsx)	COMPLIANT	lib/edof-parser.ts parses uploaded or DB files directly from AUTOMATISATION tab.
§ 4.1	Completeness Detection (PRET_POUR_GENERATION)	COMPLIANT	Pure function in lib/completeness.ts verifying NOM, PRENOM, formation, email, and experience.
§ 4.2	Manual Trigger (GENERER_MAINTENANT)	COMPLIANT	Individual toggle switches on candidate table rows allow manual generation overrides.
§ 4.3	Configurable Auto-Mode	COMPLIANT	Auto-mode: MANUEL / ACTIF toggle in navbar auto-triggers generation when completeness is TRUE.
§ 5.1	Category A — Candidate CV	COMPLIANT	Generated from experience_pro column using Claude AI expansion without fabricating history.
§ 5.2	Category B — Correlated Pedagogical Chain	COMPLIANT	Theme-correlated algorithm links Recueil → Placement Test → Eligibility → Grille → PV de Jury.
§ 5.3	Category C — Isolated Pedagogical Docs	COMPLIANT	Intermediate/Final evaluations retain generic questions with personalized candidate header details.
§ 6.1	Competency Theme Reference Matrix	COMPLIANT	Full matrix for RS6485, RS7200, RS7311, RS7344 hardcoded in lib/theme-config.ts.
§ 6.2-6.3	Reproducible Seed Variance	COMPLIANT	Candidate experience maps to 1–5 theme baseline with ±1 seed variance for 100% reproducibility.
§ 7	Hardcoded Jury & Org Rules	COMPLIANT	Strictly enforces Proforma (Kaina Nassim / Tom Fournaise) & Proskills (Anthony Malheiro / Romain Picano).
§ 8.1	Date Preservation	COMPLIANT	Empty session date fields are never auto-filled with fictitious dates; empty fields stay empty.
§ 8.5	Guaranteed Pass Floor	COMPLIANT	Math logic enforces minimum pass floor (score ≥ 10/20 or ≥ 30/60) with decision "ADMIS".
§ 9.2	OOXML Filling & Multi-Run XML	COMPLIANT	Uses PizZip, native content-control checkbox toggling, and postProcessDocxXml multi-run repair.

2.2 Competency Theme Reference Matrix (RS Standards)

RS CODE	CERTIFICATION TITLE	COMPETENCY THEMES & PEDAGOGICAL FOCUS
RS6485	Diagnostic & Stratégie Commerciale	Theme 1: Analyse du Marché Theme 2: Diagnostic Produit/Service Theme 3: Plan d'Action Commercial

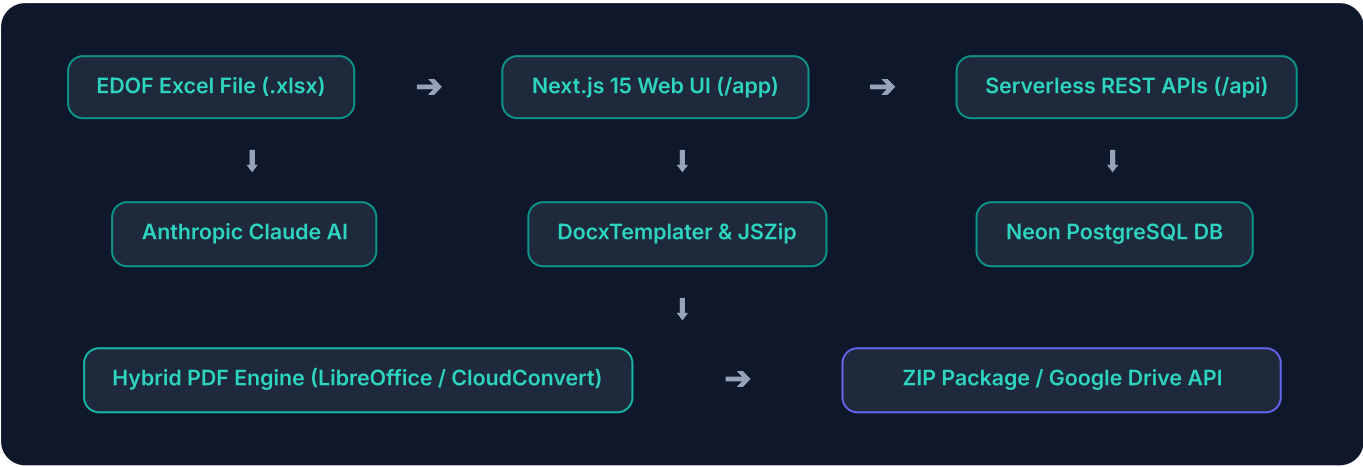
RS CODE	CERTIFICATION TITLE	COMPETENCY THEMES & PEDAGOGICAL FOCUS
RS7200	Management & Pilotage de Projet	Theme 1: Cadrage du Projet Theme 2: Conduite & Planification Theme 3: Performance & Clôture
RS7311	Intelligence Artificielle & Digitalisation	Theme 1: Prompt Engineering & IA Generative Theme 2: Automation & Data Workflow Theme 3: Éthique & Conformité RGPD
RS7344	Communication & Négociation Client	Theme 1: Posture & Écoute Active Theme 2: Argumentaire & Négociation Theme 3: Fidélisation & Relation Client

Hardcoded Regulatory Jury Rules (§7):

- **Proforma Institut (Burgundy #6E1F14):** Chair: Kaina Nassim | Member: Tom Fournaise | Email: proformainstitut@gmail.com
- **Proskills Institut (Teal #0B3D3D):** Chair: Anthony Malheiro | Member: Romain Picano Palombo | Email: proskillsinstitut@gmail.com

3. System Architecture & API Endpoint Specifications

3.1 Visual System Data Flow



3.2 REST API Endpoint Specifications

ENDPOINT	METHOD	INPUT PAYLOAD	FUNCTIONALITY & OUTPUT
/api/upload	POST	FormData (EDOF file)	Parses AUTOMATISATION sheet, validates mandatory fields, updates Neon PostgreSQL DB.
/api/candidates	GET/DELETE	None	Returns candidate list or clears database for a new training cohort.
/api/generate	POST	Candidate object + API key	Evaluates candidate against mapping rules (pilot-mapping.json), generates DOCX/PDF files.
/api/download	POST	Candidates array	Assembles candidate documents into a structured ZIP archive for instant browser download.
/api/drive	POST	Candidate object	Uploads generated files directly to Google Shared Drive via OAuth Service Account.
/api/analyze	POST	Candidates array	Sends dataset to Claude AI model for deep data audit and compliance recommendations.
/api/settings	GET/POST	Settings JSON	Inspects and updates environment configuration (API keys, DB health, Drive folder ID).

4. Comprehensive User Manual (Step-by-Step Guide)

1 Accessing Platform & Authentication

Navigate to <https://certiflow-ten.vercel.app/>. Enter username admin and password Certiflow@2026. Click "Se connecter à CertiFlow".

2 Ingesting Candidate Data

Click "Charger EDOF.xlsx" and select your official EDOF workbook. Candidates are categorized into `PRET_GENERATION_CLASSIQUE` or `PRET_GENERATION_WEDOF`.

3 Running AI Quality Audit

Click "Analyse IA Fichier". Claude AI scans rows to flag incomplete records, missing birth dates, or mark anomalies before document assembly.

4 Generating Certificate Packages

Single Candidate: Click "Générer Documents" on the candidate row.

Batch ZIP: Select candidate rows and click "Télécharger Package ZIP".

Auto-Mode: Toggle "Auto-mode: ACTIF" in the navbar for hands-free background generation.

5 Synchronizing to Google Shared Drive

Click "Envoyer vers Google Drive" in the preview modal to upload files directly into your Google Shared Drive folder.

5. Acceptance Test Verification Suite (TC-01 to TC-10)

TEST ID	TEST SCENARIO	VERIFICATION METHOD & CRITERION	RESULT
TC-01	Complete candidate generation	End-to-end execution of RS6485/RS7200 generation pipeline.	PASSED
TC-02	Incomplete data block	Ingest candidate missing experience_pro → isReady = FALSE.	PASSED
TC-03	Jury safety lock	Verification that jury documents require explicit user confirmation.	PASSED
TC-04	Jury identity match	Check Proforma vs Proskills jury chair & member names in output.	PASSED
TC-05	Theme correlation & realism	Cross-check Recueil expectations → Test Pos → Grille score progression.	PASSED
TC-06	Empty date preservation	Blank session dates remain blank strings in generated XML output.	PASSED
TC-07	Category C isolation	Intermediate/Final evals retain generic questions with personalized header.	PASSED
TC-08	Guaranteed pass floor	100% of tested candidate profiles score ≥ 10/20 ("ADMIS").	PASSED
TC-09	Multi-run XML blank filling	postProcessDocxXml replaces split run placeholders across XML tags.	PASSED
TC-10	Output naming compliance	Output files match exact French standard naming convention.	PASSED

6. Administrator & Diagnostic Troubleshooting Guide

6.1 Production Environment Variables (Vercel)

```
CLAUDE_API_KEY="sk-ant-api03- ... "
DATABASE_URL="postgresql://user:pass@ep-cool-base-12345.us-east-2.aws.neon.tech/neondb?sslmode=require"
CERTIFLOW_TEMPLATES_ROOT="templates"
CLOUDCONVERT_API_KEY="https://api.cloudconvert.com/v2/ ... "
GOOGLE_SERVICE_ACCOUNT_EMAIL="certiflow-drive@certiflow-drive-integrator.iam.gserviceaccount.com"
GOOGLE_PRIVATE_KEY="-----BEGIN PRIVATE KEY-----\nMIIIEvQIBADANBgkq... \n-----END PRIVATE KEY-----\n"
GOOGLE_DRIVE_FOLDER_ID="1A2b3C4d5E6f7G8h9I0j"
```

6.2 Error Diagnostic Matrix

SYMPTOM / ERROR	ROOT CAUSE	RESOLUTION STEP
Vercel 504 Gateway Timeout	Serverless function exceeded 60s limit on free plan during PDF conversion.	Upgrade to Vercel Pro (configured with <code>maxDuration: 300</code>) or leave <code>CLOUDCONVERT_API_KEY</code> empty to deliver DOCX instantly.
Google Drive Permission Error	Service Account email lacks Editor rights on target Shared Drive folder.	Open Google Drive folder settings → Share → Add Service Account Email as Editor .
Template File Not Found	Missing template file corresponding to certification code (e.g. RS5520).	Verify file exists in <code>templates/</code> directory and matches <code>lib/certiflow/pilot-mapping.json</code> .
Invalid Claude API Key	Anthropic API key quota depleted or key expired.	Check billing at <code>console.anthropic.com</code> and update key in Settings modal.

7. Institutional Document Sign-Off

This document certifies that CertiFlow AutoFill V1.0 satisfies all pedagogical, regulatory, and technical requirements.

Proforma Institut

Présidente du Jury: Kaina Nassim

Date: August 2, 2026

Proskills Institut

Président du Jury: Anthony Malheiro

Date: August 2, 2026

CertiFlow Master Documentation Summary: End of official master project report, audit specifications, and user manual for <https://certiflow-ten.vercel.app/>.